Zilionis et al. 2019 — Non-Small Cell Lung Cancer

inDrop · 3,920 malignant cells (6/7 samples; mixed LUSC + LUAD histology) · developmental-program cell-state analysis

Geometry: LINEAR (non-branching) Gate 1 GO · Gate 2 GO · Gate 3 NA

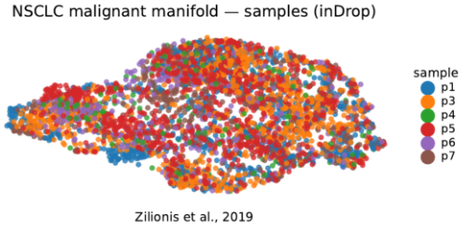
GO / NO-GO GATES

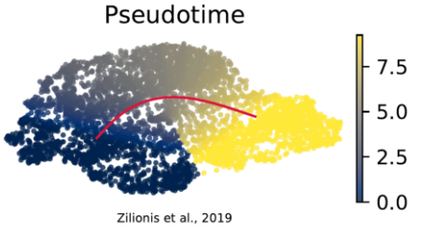
Gate 1 — integration & continuity: **GO**
All 6 samples and both histologies intermixed on ONE connected manifold (sample iLISI 3.4/6 at theta=10; 0 detached islands). Mixed + continuous — judged on the embedding, not a scalar.

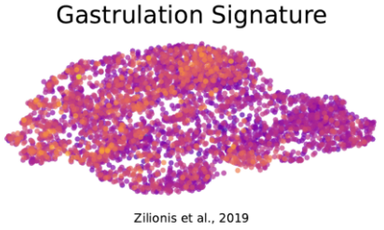
Gate 2 — module geography: **GO (non-branching)**
Shared developmental gradient: monotone Gast(origin, rho -0.59) -> pSom(terminus, rho +0.76) axis; Crest an inverse off-axis readout. Present but collapsed onto one axis, not two competing arms.

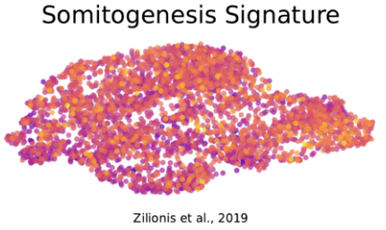
Gate 3 — hub cross-repression: **NA**
No Gast/pSom co-expression HUB (co-occurrence enrichment 0.07 smoothed / 0.70 raw, both <1) and no stable branch point (transcriptome PPT = 2 tips/0 forks across nodes x seeds). Non-branching -> NA.

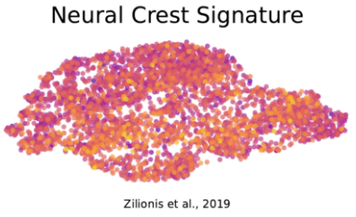
STEP FIGURES

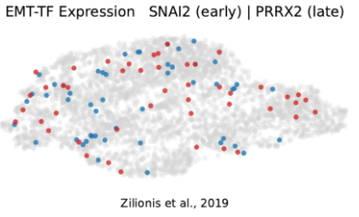












PARAMETER CHOICES & RATIONALE

- Threshold: >=131 malignant (geom-mean across the 232->75 count cliff, ratio 3.09) -> drops p2 (75) -> 3,920 cells.
- Overdispersed genes 248 (scFates GAM); PCA 50 PC, n_pcs=20 (~50% variance; droplet scale).
- Integration: Harmony on sample, theta=10 (iLISI plateau 1.4->3.4/6); n_neighbors=30; UMAP spread=1.3. 0 islands.
- Trajectory: scFates CURVE (ElPiGraph) Nodes=20, epg_lambda=0.3, epg_mu=100 on UMAP; 2 tips/0 forks, fold-free (self-approach 2.45); continuity corr(pt,DC1)=-0.85 (independent embedding).
- Root: Gast-high tip (gastrulation = earliest developmental program; Crest is late, never the root).
- EMT-TFs (detection-gated, VIM excluded; FOXC/PRRX included): SNAI2 (early/root) <-> PRRX2 (late/terminus); weak, root-loaded.

VERDICT: The embryonic pattern reproduces on a single collapsed axis — a gastrulation-high origin resolving to a somitogenesis-high terminus, with neural crest as an inverse off-axis gradient (flagged: non-terminal). Not the Puram hub-and-spoke bifurcation.